Grand Unification in Recognition Science: One Ledger, Three Sectors

Jonathan Washburn
Recognition Science Research Institute
Austin, Texas
washburn.jonathan@gmail.com

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Abstract

We present the Recognition Science approach to grand unification. Unlike conventional GUTs (which embed $SU(3) \times SU(2) \times U(1)$ into a larger group like $SU(5)$ or $SO(10)$), RS unification is structural: all three gauge sectors originate from a single recognition ledger. The three sectors are different topological projections of the same D-cube ($D = 3$): color from face-pairs, weak isospin from spinors, hypercharge from overall phase. This explains why the couplings are related without requiring a GUT scale or predicting proton decay. The "unification scale" is not a high-energy threshold but the ledger itself—already unified at all scales. All structural results follow from the RS forcing chain [8].

1 Introduction

Grand unification attempts to explain why the three gauge couplings appear to converge at high energies [2]. Georgi and Glashow [2] proposed $SU(5)$ as the minimal GUT: fermions in $\mathbf{5}^* \oplus \mathbf{10}$, with $M_{\text{GUT}} \sim 10^{16}$ GeV from RG running. Pati and Salam [4] extended via $SU(4) \times SU(2)_L \times SU(2)_R$; Georgi [3] and Fritzsch–Minkowski [5] proposed $SO(10)$ with a $\mathbf{16}$ spinor. All require a GUT scale and predict proton decay and monopoles.

RS alternative. Recognition Science [8] (RS) offers a different kind of unification. The cost functional $J(x) = \frac{1}{2}(x + x^{-1}) - 1$ and the golden ratio $\varphi = (1 + \sqrt{5})/2$ force $D = 3$ spatial dimensions. The D-cube geometry then forces $SU(3) \times SU(2) \times U(1)$: one ledger, three sectors as different projections. No GUT scale is needed—the structure is already unified at all scales. No proton decay or monopoles are predicted.

2 Recognition Science Framework

Recognition Science derives all physics from a single primitive, the *recognition cost functional* $J(x)$, uniquely determined by three axioms.

Definition 2.1 (RS Cost Functional)**.** For $x > 0$:

$$J(x) = \frac{1}{2}\big(x + x^{-1}\big) - 1. \tag{1}$$

Axiom A1 (Normalization): $J(1) = 0$.
Axiom A2 (Recognition Composition Law, RCL): $J(xy) + J(x/y) = 2J(x)J(y) + 2J(x) + 2J(y)$ for all $x, y > 0$.
Axiom A3 (Calibration): $J''_{\log}(0) = 1$, where $J_{\log}(t) := J(e^t)$.

Theorem 2.2 (Cost Uniqueness)**.** *The unique continuous solution to* (A1)–(A3) *is* $J(x) = \frac{1}{2}(x + x^{-1}) - 1$.

Proof sketch. Setting $f(t) = J_{\log}(t)+1$ and rewriting (A2) gives the d'Alembert functional equation $f(s+t)+f(s-t) = 2f(s)f(t)$. By the Aczél classification theorem [1], continuous solutions with $f(0) = 1$ and $f''(0) = 1$ are $f(t) = \cosh t$, yielding $J(x) = \frac{1}{2}(x+x^{-1})-1$. □

The golden ratio $\varphi = (1 + \sqrt{5})/2$ is forced by self-similarity (the φ-ladder). Spatial dimension $D = 3$ is forced by the gap-45 synchronization $\mathrm{lcm}(8, 45) = 360$. The $D = 3$ cube structure is the geometric foundation of all three gauge sectors.

3 Structural Unification

Proposition 3.1 (Single Ledger — Structural Argument)**.** *All three gauge sectors (strong, weak, electromagnetic) originate from a single recognition ledger. The apparent multiplicity is a projection effect.*

Structural argument. In RS, the entire physical state is encoded in the recognition ledger $L = \{(x_i, J(x_i))\}$, a single double-entry accounting system. The three gauge symmetries are not separate structures imposed on the ledger but are consequences of the three independent structural features of the $D = 3$ cube (Theorem 4.3): face-pairs, spinor lifts, and J-cost phase. Since all three derive from the same cube geometry and cost functional, they share one common origin. There is no separate "strong sector ledger" or "weak sector ledger"—only the three projections of one structure. □

Proposition 3.2 (Common Origin from $D = 3$ — Structural Argument)**.** $N_c = 3$ *(strong),* $N_{\text{gen}} = 3$ *(weak generations), and the* $U(1)$ *hypercharge all follow from the single structural input* $D = 3$.

Structural argument. From Lemma 4.2: $\binom{3}{2} = 3$ independent coordinate planes $\Rightarrow N_c = 3$ color channels ($SU(3)$). The $SO(3)$ spinor representation is a doublet of $SU(2)$, giving two weak isospin states. Three colors times two isospin states times the overall J-cost phase yields one generation. The φ-ladder sectors before first periodicity of the recognition angle $\theta_0 = \arccos(1/4)$ give three distinct sectors, motivating $N_{\text{gen}} = 3$; however, the rigorous derivation of exactly three generations is an identified open problem in RS (see Remark below). □

Remark 3.3 (Open problem: three generations). The argument that $\theta_0 = \arccos(1/4)$ forces exactly three φ-rung sectors before periodicity is a structural motivation, not a rigorous derivation. A complete proof would require showing that the φ-ladder restricted to the recognition angle has exactly three periodic sectors and that each sector corresponds to a physical generation of fermions. This is one of the most important open problems in the RS framework.

4 Three Sectors as Projections of One D-Cube

Definition 4.1 (D-Cube)**.** The D-cube has vertices $\{0,1\}^D$. For $D = 3$: 8 vertices, 12 edges, 6 faces (opposite-face pairs: 3), and 3 independent coordinate planes (2-plane pairs: $\binom{3}{2} = 3$). For $D = 3$ specifically, the number of opposite-face pairs equals the number of 2-plane pairs ($D = \binom{D}{2} = 3$); this coincidence holds only at $D = 3$.

Lemma 4.2 (2-Plane Count for $D = 3$)**.** *The $D = 3$ cube has $\binom{3}{2} = 3$ coordinate planes: (xy), (xz), (yz). These define 3 independent complex recognition channels.*

Proposition 4.3 (Projection Structure — Structural Argument)**.** *The three gauge sectors are structural projections of the $D = 3$ cube: (i)* ***Color*** *($SU(3)$): the 3 coordinate planes each index a complex recognition channel, motivating $SU(3)$; (ii)* ***Weak*** *($SU(2)$): the $SO(3)$ rotation group of the 3-cube has a doublet spinor representation lifting to $SU(2)$; (iii)* ***Hypercharge*** *($U(1)$): $J(x) = \frac{1}{2}(x + x^{-1}) - 1$ is invariant under phase rotation $x \mapsto e^{i\theta}x$, giving $U(1)$.*

Remark 4.4 (Status of the gauge group derivation). The identifications (i)–(iii) are structural: the face-pair count 3 corresponds to $SU(3)$; the $SO(3)$ spinor lift gives $SU(2)$; the J-phase symmetry gives $U(1)$. The gap in each case is showing that the *representation content* (which irreducible representations the quarks and leptons transform under) follows from the RS ledger without additional assumptions. This is an identified open problem.

Corollary 4.5. *RS unification does not embed $SU(3) \times SU(2) \times U(1)$ into a larger group. The three sectors are already facets of one structure.*

5 Gauge Coupling Relations Without Running

Conventional GUTs relate couplings by renormalization-group running to a common scale M_{GUT}. RS relates them by geometry at all scales.

Proposition 5.1 (Coupling Relations from $D = 3$ — Summary)**.** *The gauge couplings are determined by the D-cube structure with zero free parameters:*

- $\sin^2\theta_W = (3 - \varphi)/6 \approx 0.230$ *from the electroweak φ-structure;*
- $\alpha_s = 2/17 \approx 0.1176$ *at M_Z from the 17 2D crystallographic (wallpaper) groups;*
- $\alpha^{-1} \approx 137$ *from the cube passive-edge count and f_{gap} correction.*

Each derivation is developed in full in companion RS papers on the Weinberg angle, strong coupling, and fine-structure constant, respectively [8].

Remark 5.2 (Scale independence and falsifiability). The coupling values are set geometrically by $D = 3$ and hold at all scales without RG running. This is a sharp departure from conventional GUTs: the RS values (e.g. $\sin^2\theta_W = (3 - \varphi)/6 \approx 0.2306$) should be *scale-independent*, while measured couplings run with energy. If precision measurements establish significant running in $\sin^2\theta_W$ below M_Z, the RS geometric derivation would be under pressure.

6 No Proton Decay

The Georgi–Glashow $SU(5)$ GUT predicts proton decay via X/Y leptoquark bosons mediating $q \leftrightarrow \ell$, e.g. $p \to e^+\pi^0$. Predicted $\tau_p \sim 10^{30}$–10^{35} yr; experiments give $\tau_p > 10^{34}$ yr, ruling out minimal $SU(5)$.

Theorem 6.1 (No Proton Decay in RS)**.** *RS does not predict proton decay.*

Proof. No embedding into $SU(5)$ or $SO(10) \Rightarrow$ no X/Y bosons. The three sectors are facets of one ledger, not broken remnants. No gauge bosons mediate $q \leftrightarrow \ell$. □

7 No GUT Monopoles

When $SU(5)$ breaks to $SU(3) \times SU(2) \times U(1)$, the vacuum manifold has $\pi_2(G/H) \neq 0$, so monopoles are topologically required (t'Hooft–Polyakov [6, 7]), with mass $\sim M_{\mathrm{GUT}}/\alpha \sim 10^{16}$ GeV.

Theorem 7.1 (No GUT Monopoles)**.** *RS does not predict grand-unification monopoles.*

Proof. Monopoles require $G \to H$ with $\pi_2(G/H) \neq 0$. In RS there is no such breaking; $SU(3) \times SU(2) \times U(1)$ is the direct product of projections, not a broken remnant. π_2 is trivial. □

8 Comparison with Conventional GUTs

	Conventional GUT	**RS**
Unification	Larger group ($SU(5)$, $SO(10)$)	Single ledger
GUT scale	$\sim 10^{16}$ GeV (RG running)	None (all scales)
Proton decay	Predicted (X/Y bosons)	Not predicted
Monopoles	Topologically required	Not predicted
Couplings	Run to M_{GUT}	From D-cube geometry
Parameters	M_{GUT}, breaking sector	Zero

9 Predictions and Falsifiers

P1. **P1**: No proton decay. Falsifier: $p \to e^+\pi^0$ or similar.

P2. **P2**: No GUT monopoles. Falsifier: monopoles $\sim 10^{16}$ GeV.

P3. **P3**: $\sin^2\theta_W = (3-\varphi)/6$, $\alpha_s = 2/17$, $\alpha^{-1} \approx 137$. Falsifier: measurements outside bands.

P4. **P4**: Exactly three generations. Falsifier: fourth light generation.

P5. **P5**: No X/Y bosons. Falsifier: $q \leftrightarrow \ell$ via new gauge bosons.

10 Conclusion

Grand unification in RS is structural: one ledger yields three gauge sectors as different projections of the D-cube. No GUT scale, no proton decay, no monopoles. The coupling relations derive from $D = 3$ geometry, not from renormalization-group running. RS offers a falsifiable alternative to conventional GUTs, with distinct experimental signatures.

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